**Introduction to Data Science**
**Fall - 2025**
**Assignment- 3**
**Issue Date: 24th October 2025**
**Submission Deadline: 30th November 2025**
**Group Submission - Max 3 allowed**

**Deliverables: Submit a short report with Jupyter Notebook.**

**Evaluation Criteria: Viva about Model training steps,Evaluation metric outputs with interpretations of results, Graphs (residual plots, ROC curve)**

**TASK-1:**

Implement two versions of a simple k-means algorithm: one of which uses numerical centroids as centers, the other of which restricts centers to be input points from the dataset. Then experiment. Which algorithm converges faster on average? Which algorithm produces clusterings with lower absolute and mean-squared error and by how much?

**TASK-2:**

Identify a dataset on entities where you have some sense of natural clusters which should emerge, be it on people, universities, companies, or movies. Implement Clustering using k-means and agglomerative algorithms. Evaluate the resulting clusters based on your knowledge of the domain. Which algorithm did a good job? What things did it get wrong? Do experiments studying the impact of merging criteria(single link, complete link, average link) on the properties of resulting cluster tree. Which leads to the tallest trees? Which method produces results most consistent with k-MEANS clustering?

**TASK-3:**

Build a Linear Regression model to predict the Closing Gold Price (Close) using other numerical features such as Open, High, Low, or USD/INR.What is the value of $R^2$, and how well does the model fit the data?Does the residual plot show any visible patterns (non-linearity or bias)?Which feature has the strongest correlation with gold price?

Dataset Name : Gold Price Data (1979–2021)

https://www.kaggle.com/datasets/altruistdelhite04/gold-price-data

**TASK-4:Attempt any 2 challenges and submit your score with a link too.**

1. Which people are most influential in a given social network?
   https://www.kaggle.com/c/predict-who-is-more-influential-in-a-social-network
2. Predict which product a consumer is most likely to buy.
   https://www.kaggle.com/c/coupon-purchase-prediction
3. Identify what is being cooked, given the list of ingredients.
   https://www.kaggle.com/c/whats-cooking
4. Decide whether a particular student will answer a given question correctly
   https://www.kaggle.com./c/WhatDoYouKnow